

Báo cáo bài tập tuần 1

Hoàng Đức Anh – 20235640

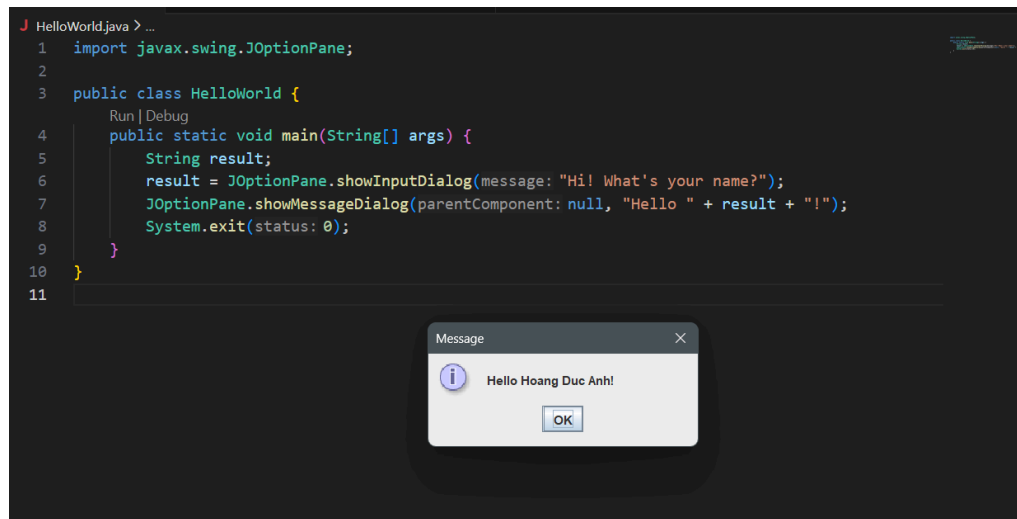
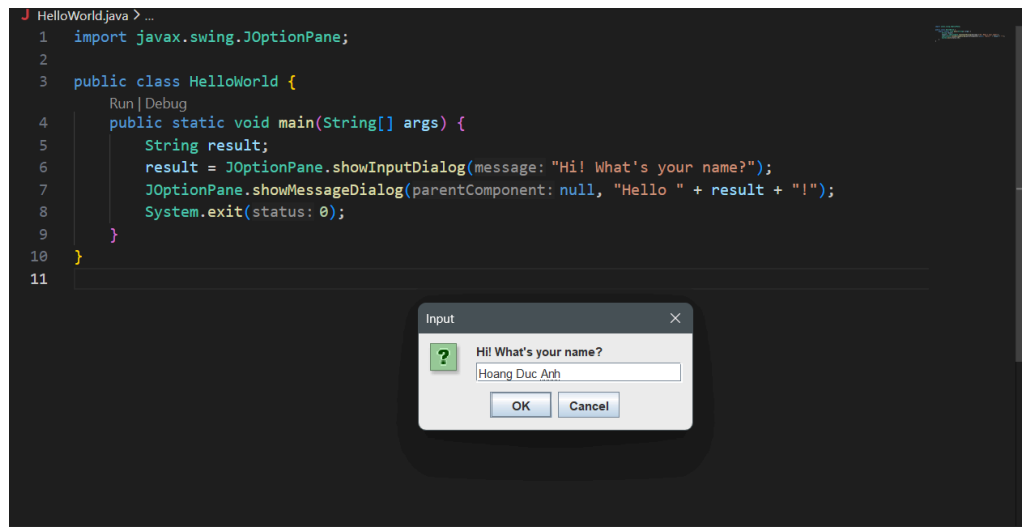
1. HelloWorld

a) Code :

```
import javax.swing.JOptionPane;

public class HelloWorld {
    public static void main(String[] args) {
        String result;
        result = JOptionPane.showInputDialog("Hi!
What's your name?");
        JOptionPane.showMessageDialog(null, "Hello "
+ result + "!");
        System.exit(0);
    }
}
```

b) Kết quả:



2. Calculator build by Swing

a) Code:

- CalculatorController:

```
package MVCSwing;

import java.awt.event.*;

public class CalculatorController {

    private CalculatorModel model;
    private CalculatorView view;

    public CalculatorController(CalculatorModel model, CalculatorView view) {
        this.model = model;
    }
}
```

```

        this.view = view;

        view.btnCalculate.addActionListener(new ActionListener() {
            public void actionPerformed(ActionEvent e) {
                try {
                    int a = Integer.parseInt(view.txtA.getText());
                    int b = Integer.parseInt(view.txtB.getText());
                    String op = (String) view.cbOperator.getSelectedItem();

                    double result = model.calculate(a, b, op);
                    view.lblResult.setText("Result: " + result);

                } catch (NumberFormatException ex) {
                    view.lblResult.setText("You need to enter 2 integers");
                } catch (Exception ex) {
                    view.lblResult.setText("Error!");
                }
            }
        });
    }
}

```

- CalculatorMode:

```

package MVCSwing;

public class CalculatorModel {

    public double calculate(double a, double b, String op) {
        switch (op) {
            case "+": return a + b;
            case "-": return a - b;
            case "*": return a * b;
            case "/":
                if (b != 0) return a / b;
                else throw new ArithmeticException("Cannot divide by 0");
            default: return 0;
        }
    }
}

```

- CalculatorView:

```

package MVCSwing;

import javax.swing.*;
import java.awt.*;

public class CalculatorView extends JFrame {

```

```

    JTextField txtA = new JTextField(5);
    JTextField txtB = new JTextField(5);
    JComboBox<String> cbOperator = new JComboBox<>(new String[]{"+", "-",
    "*", "/" });
    JButton btnCalculate = new JButton("Calculate");
    JLabel lblResult = new JLabel("Result: ");

    public CalculatorView() {
        setTitle("Calculator - Swing MVC");
        setLayout(new FlowLayout());
        setSize(300,150);
        setDefaultCloseOperation(EXIT_ON_CLOSE);

        add(txtA);
        add(cbOperator);
        add(txtB);
        add(btnCalculate);
        add(lblResult);

        setVisible(true);
    }
}

```

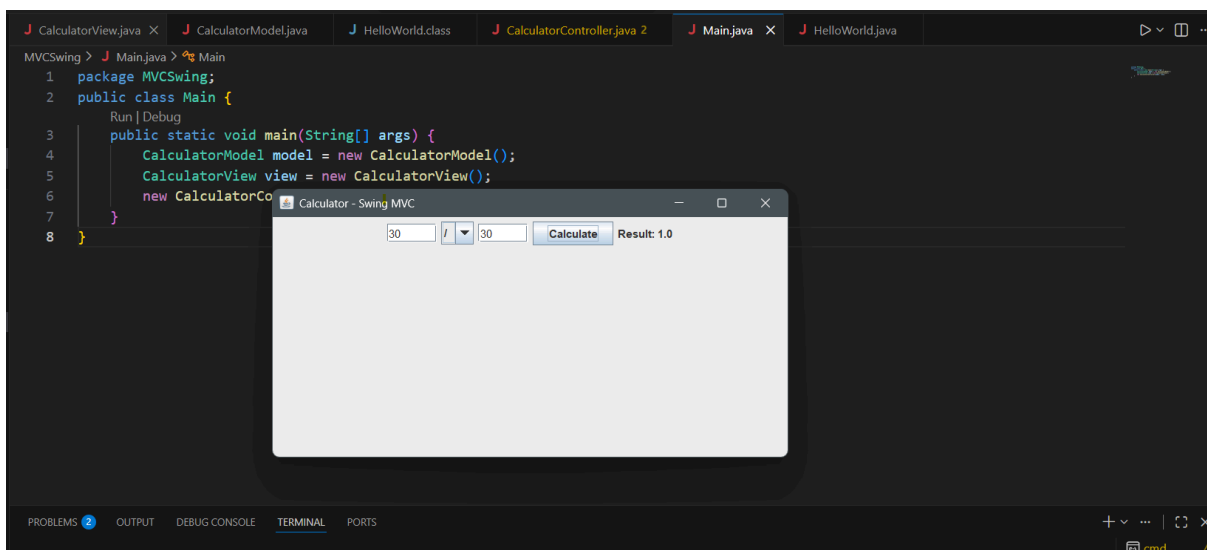
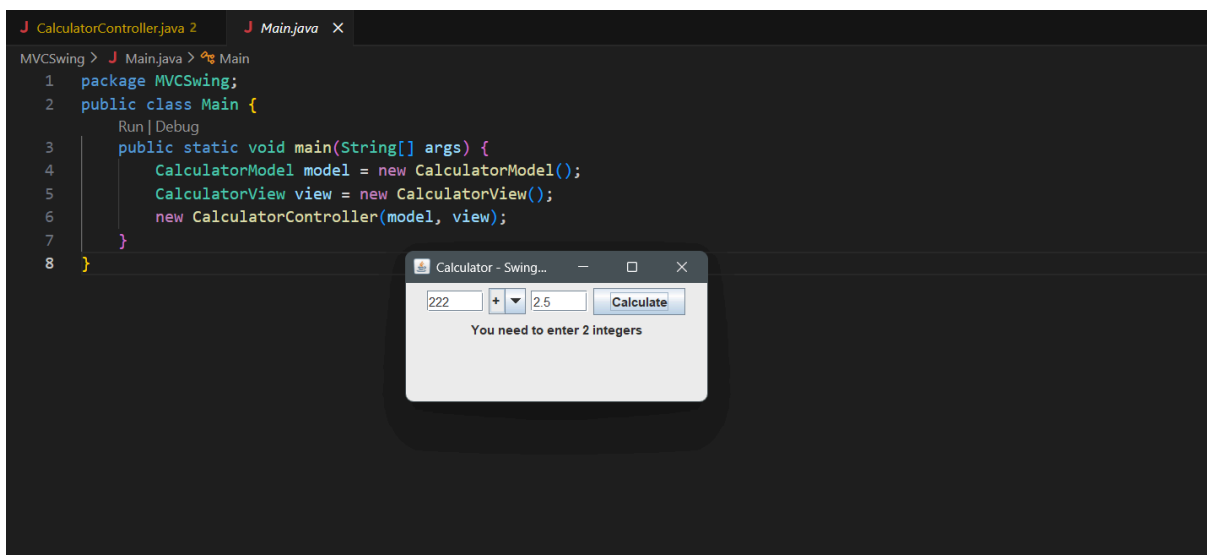
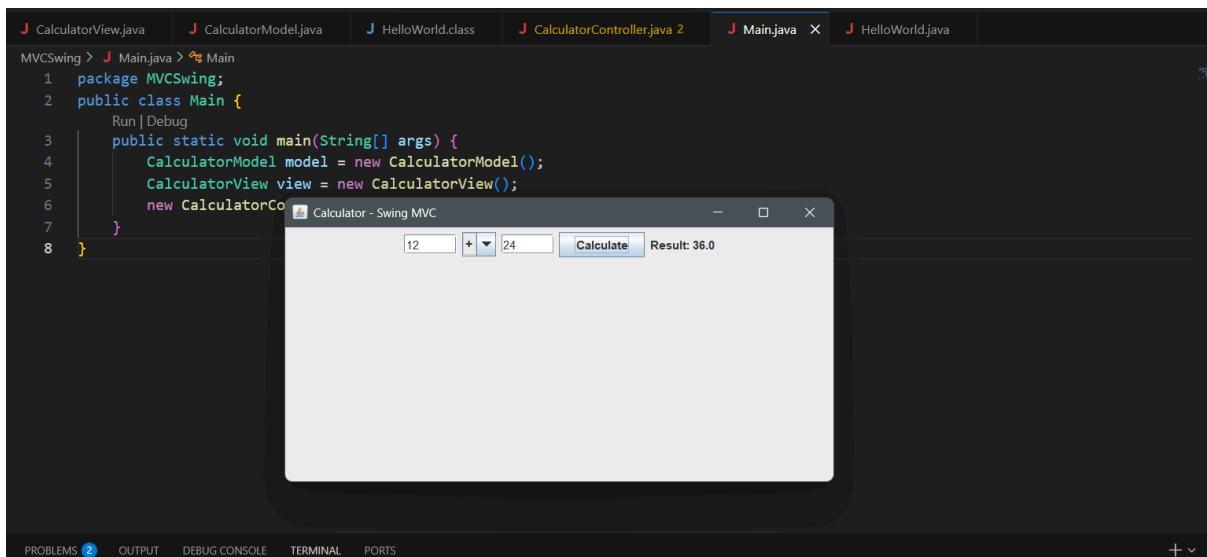
- Main:

```

package MVCSwing;
public class Main {
    public static void main(String[] args) {
        CalculatorModel model = new CalculatorModel();
        CalculatorView view = new CalculatorView();
        new CalculatorController(model, view);
    }
}

```

b) Kết quả:



3. Calculator build by JavaFX

4. Code:

- Calculator.fxml

```
<?xml version="1.0" encoding="UTF-8"?>

<?import javafx.scene.control.*?>
<?import javafx.scene.layout.*?>

<VBox spacing="10" alignment="CENTER"
      xmlns="http://javafx.com/javafx/17"
      xmlns:fx="http://javafx.com/fxml"
      fx:controller="CalculatorController">

    <Label text="Calculator MVC" />

    <TextField fx:id="txtA" promptText="Enter first integer"/>

    <ComboBox fx:id="cbOperator"/>

    <TextField fx:id="txtB" promptText="Enter second integer"/>

    <Button text="Calculate" onAction="#handleCalculate"/>

    <Label fx:id="lblResult" text="Result:" />

</VBox>
```

- CalculatorController

```
package MVCJavaFX;

import javafx.fxml.FXML;
import javafx.scene.control.*;

public class CalculatorController {

    @FXML
    private TextField txtA;

    @FXML
    private TextField txtB;

    @FXML
```

```

private ComboBox<String> cbOperator;

@FXML
private Label lblResult;

private CalculatorModel model = new CalculatorModel();

@FXML
public void initialize() {
    cbOperator.getItems().addAll("+", "-", "*", "/");
    cbOperator.setValue("+");
}

@FXML
private void handleCalculate() {

    try {
        int a = Integer.parseInt(txtA.getText());
        int b = Integer.parseInt(txtB.getText());
        String op = cbOperator.getValue();

        double result = model.calculate(a, b, op);
        lblResult.setText("Result: " + result);

    } catch (NumberFormatException e) {
        lblResult.setText("You need to enter 2 integers");
    } catch (ArithmeticException e) {
        lblResult.setText("Cannot divide by zero");
    } catch (Exception e) {
        lblResult.setText("Error!");
    }

}
}

```

- CalculatorModel

```

package MVCJavaFX;

public class CalculatorModel {

```

```

    public double calculate(int a, int b, String operator) {

        switch (operator) {
            case "+":
                return a + b;
            case "-":
                return a - b;
            case "*":
                return a * b;
            case "/":
                if (b == 0) {
                    throw new ArithmeticException("Divide by
zero");
                }
                return (double) a / b;
            default:
                throw new IllegalArgumentException("Invalid
operator");
        }
    }
}

```

- Main

```

package MVCJavaFX;

import javafx.application.Application;
import javafx.fxml.FXMLLoader;
import javafx.scene.Scene;
import javafx.stage.Stage;

public class Main extends Application {

    @Override
    public void start(Stage stage) throws Exception {

        FXMLLoader loader = new FXMLLoader(
            getClass().getResource("calculator.fxml"));
    }
}

```

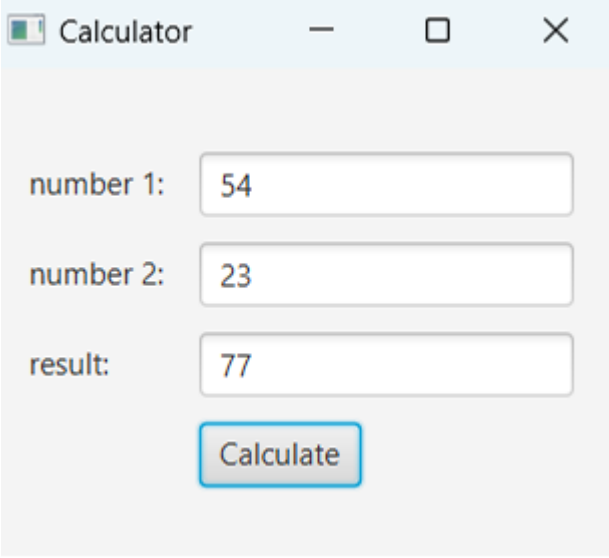


```
Scene scene = new Scene(loader.load());

stage.setTitle("Calculator MVC");
stage.setScene(scene);
stage.show();
}

public static void main(String[] args) {
    launch();
}
}
```

5. Kết quả:

A screenshot of a Java Swing window titled "Calculator". It contains three text input fields. The first is labeled "number 1:" and contains the value "54". The second is labeled "number 2:" and contains the value "23". The third is labeled "result:" and contains the value "77". Below the input fields is a button labeled "Calculate".

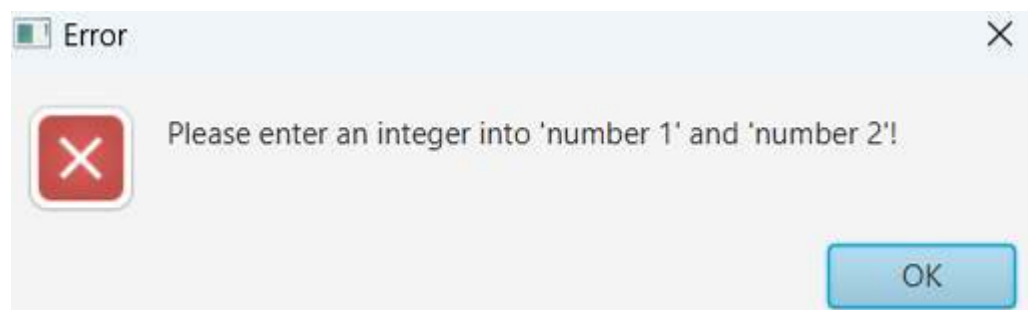
Calculator

number 1: 54

number 2: 23

result: 77

Calculate



4. Git

a) Add

```
c:\Users\hoang\Downloads\GIT\2025.2-166155-06\Homework01\20235640-HoangDucAnh>git add .

c:\Users\hoang\Downloads\GIT\2025.2-166155-06\Homework01\20235640-HoangDucAnh>git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   HelloWorld.class
    modified:   HelloWorld.java
```

b) Remove

```
C:\Users\hoang\Downloads\GIT\2025.2-166155-06\Homework01\20235640-HoangDucAnh>git rm README.txt
rm 'Homework01/20235640-HoangDucAnh/README.txt'

C:\Users\hoang\Downloads\GIT\2025.2-166155-06\Homework01\20235640-HoangDucAnh>git commit -m "Delete readme"
[main 0b51a22] Delete readme
 1 file changed, 1 deletion(-)
 delete mode 100644 Homework01/20235640-HoangDucAnh/README.txt

C:\Users\hoang\Downloads\GIT\2025.2-166155-06\Homework01\20235640-HoangDucAnh>git push
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 4 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (4/4), 339 bytes | 169.00 KiB/s, done.
Total 4 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To https://github.com/ryujin05/2025.2-166155-06.git
 68b7c67..0b51a22  main -> main
```

c) Modify

```
C:\Users\hoang\Downloads\GIT\2025.2-166155-06\Homework01\20235640-HoangDucAnh>echo Hello > test.txt

C:\Users\hoang\Downloads\GIT\2025.2-166155-06\Homework01\20235640-HoangDucAnh>git add test.txt

C:\Users\hoang\Downloads\GIT\2025.2-166155-06\Homework01\20235640-HoangDucAnh>git add test.txt

C:\Users\hoang\Downloads\GIT\2025.2-166155-06\Homework01\20235640-HoangDucAnh>git commit -m "add test.txt"
[main d982cf8] add test.txt
 1 file changed, 1 insertion(+)
 create mode 100644 Homework01/20235640-HoangDucAnh/test.txt

C:\Users\hoang\Downloads\GIT\2025.2-166155-06\Homework01\20235640-HoangDucAnh>echo Hello Git > test.txt

C:\Users\hoang\Downloads\GIT\2025.2-166155-06\Homework01\20235640-HoangDucAnh>git status
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
    modified:   test.txt
```

d) Commit

```
c:\Users\hoang\Downloads\GIT\2025.2-166155-06\Homework01\20235640-HoangDucAnh>git commit -m "add 1st exercise"
[main 6b63ec4] add 1st exercise
 2 files changed, 10 insertions(+), 1 deletion(-)
 create mode 100644 Homework01/20235640-HoangDucAnh/HelloWorld.class
```

e) Push

```
c:\Users\hoang\Downloads\GIT\2025.2-166155-06\Homework01\20235640-HoangDucAnh>git push -u origin main
Enumerating objects: 10, done.
Counting objects: 100% (10/10), done.
Delta compression using up to 4 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (6/6), 1.16 KiB | 596.00 KiB/s, done.
Total 6 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/ryujin05/2025.2-166155-06.git
   ef03c89..6b63ec4  main -> main
branch 'main' set up to track 'origin/main'.
```

f) Pull

```
c:\Users\hoang\Downloads\GIT\2025.2-166155-06\Homework01>git pull
remote: Enumerating objects: 81, done.
remote: Counting objects: 100% (80/80), done.
remote: Compressing objects: 100% (45/45), done.
remote: Total 75 (delta 25), reused 66 (delta 19), pack-reused 0 (from 0)
Unpacking objects: 100% (75/75), 554.26 KiB | 299.00 KiB/s, done.
From https://github.com/ryujin05/2025.2-166155-06
   6b63ec4..68b7c67  main       -> origin/main
* [new branch]      Liem       -> origin/Liem
* [new branch]      Tdt        -> origin/Tdt
* [new branch]      hw1-hhoang -> origin/hw1-hhoang
Updating 6b63ec4..68b7c67
Fast-forward
 Homework01/20235729-PhanHuyHoang/.gitignore | 1 +
 .../20235729-PhanHuyHoang/.vscode/launch.json | 31 ++++++
 .../20235729-PhanHuyHoang/.vscode/settings.json | 5 ++
 Homework01/20235729-PhanHuyHoang/Hw01-ITSS.pdf | Bin 0 -> 200416 bytes
 Homework01/20235729-PhanHuyHoang/HelloWorld.java | 15 +++
 .../MVC/javafx/CalculatorController.java | 34 ++++++
 .../MVC/javafx/CalculatorModel.java | 32 ++++++
 .../MVC/javafx/CalculatorView.java | 80 ++++++
 .../MVC/javafx/MVCCalculator.java | 25 +++++
 .../MVC/swing/CalculatorController.java | 43 ++++++
 .../MVC/swing/CalculatorModel.java | 32 ++++++
 .../MVC/swing/CalculatorView.java | 68 ++++++
 12 files changed, 286 insertions(+), 1 deletion(-)
 create mode 100644 Homework01/20235729-PhanHuyHoang/Hw01-ITSS.pdf
```

g) Merge

merge now #6

 **Merged** hoangducanh2005 merged 1 commit into [main](#) from [HDA](#)  now

 Conversation **1**  Commits **1**  Checks **0**  Files changed **1**



hoangducanh2005 commented [8 minutes ago](#) ...

No description provided.



 up 8a77c0c



hoangducanh2005 commented [6 minutes ago](#) Author ...

oke check



 hoangducanh2005 merged commit [01e3f8c](#) into [main](#) [now](#) Revert

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h) Branch

```
C:\Users\hoang\Downloads\GIT\2025.2-166155-06>git branch HDA
```

```
C:\Users\hoang\Downloads\GIT\2025.2-166155-06>git checkout HDA
M      Homework01/20235640-HoangDucAnh/HelloWorld.java
Switched to branch 'HDA'
```

```
C:\Users\hoang\Downloads\GIT\2025.2-166155-06>
```

```
c:\Users\hoang\Downloads\GIT\2025.2-166155-06>git push -u origin HDA
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'HDA' on GitHub by visiting:
remote:   https://github.com/ryujin05/2025.2-166155-06/pull/new/HDA
remote:
To https://github.com/ryujin05/2025.2-166155-06.git
 * [new branch]      HDA -> HDA
branch 'HDA' set up to track 'origin/HDA'.
```